

Structural Bio-AI

From Sequence to Structure

How Machine Learning Cracked Biology's 50-Year Grand Challenge

Klara Kropivšek Brumat, PhD | University of Nova Gorica | April 3, 2026



SMASH
machine learning for science and humanities postdoctoral program

What We'll Cover Today

1

Biology Foundations

Proteins, folding problem, classical methods, Nobel Prizes

1:00–1:30

2

The ML Revolution

CNNs → Transformers → AlphaFold architecture

1:30–2:30

3

Critical Evaluation

Benchmarking: where models fail (our research)

2:50–3:15

4

Case Study

Practical guide, diagnostic exercise, takeaways

3:15–4:00

 Break: 2:30–2:50

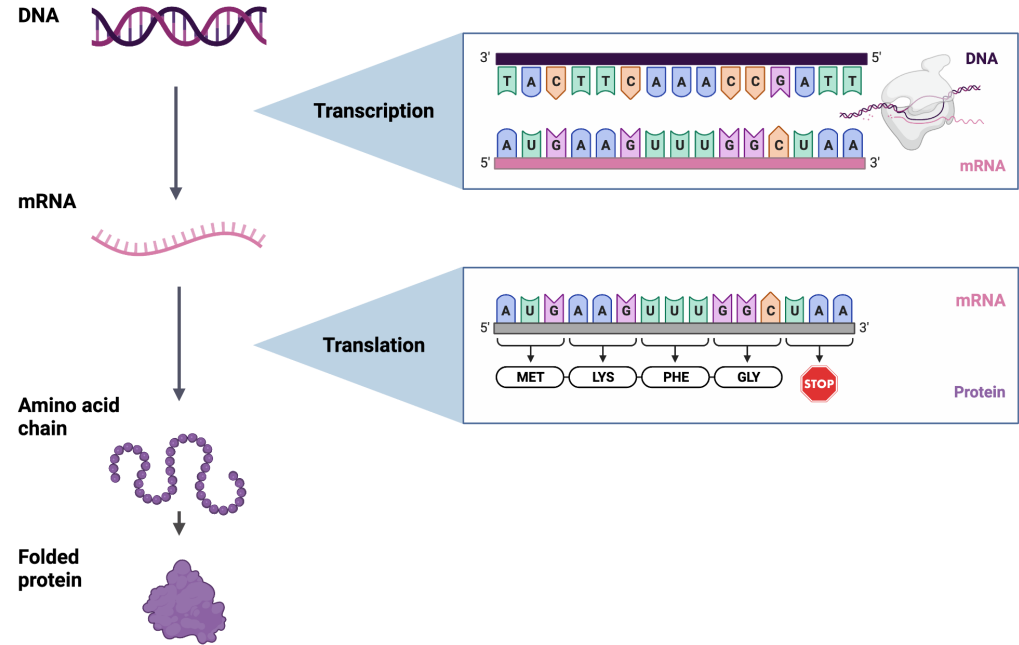
Part 1: The Biological Grand Challenge

What is a protein? Why does shape matter? How did we solve structures before AI?

1:00 – 1:30 PM

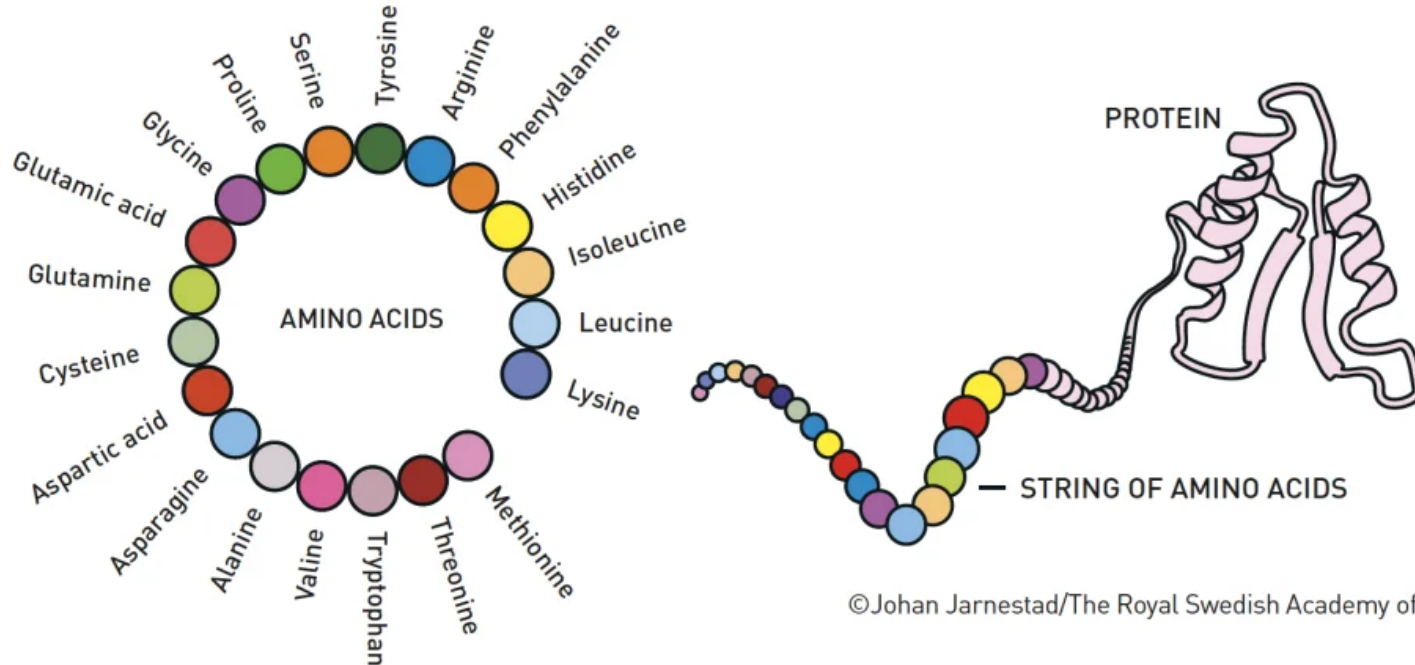
Biology is Digital: The Code of Life

- **DNA** = the hard drive (stores instructions)
- **RNA** = the messenger (temporary copy)
- **Proteins** = the hardware — molecular machines that do the actual work
- Proteins carry oxygen, contract muscles, fight viruses, catalyze reactions
- Key insight: proteins start as simple 1D strings of information → must fold into 3D



Building Blocks: The 20-Letter Alphabet

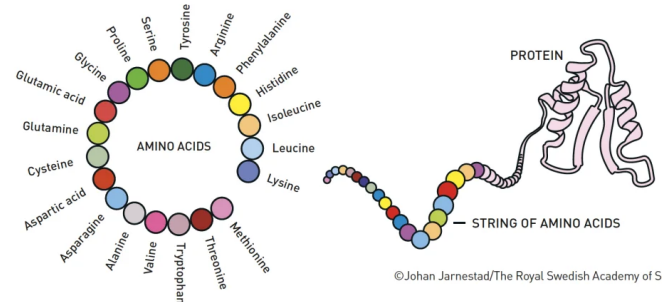
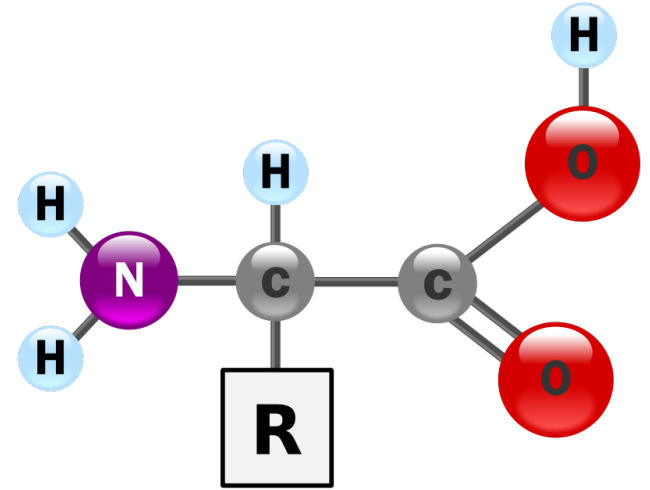
Proteins are chains of amino acids (residues) — typically 100 to 1000 long



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Building Blocks: The 20-Letter Alphabet

- Proteins are chains of amino acids (residues) — typically 100 to 1000 long
- All share the same backbone (N-C α -C), but differ in side chain ('R group')
- 20 varieties: oily (hydrophobic), charged, bulky, tiny
- Think LEGO: same connectors, 20 different shapes sticking out the side
- Side chains push, pull, interact → the chain folds into a specific 3D shape

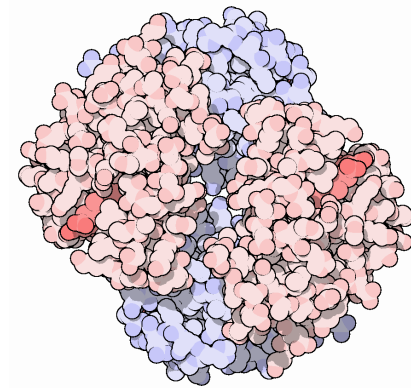


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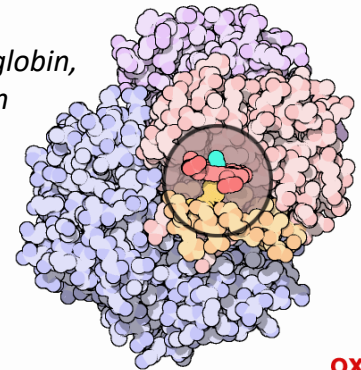
Structure = Function: Shape Is Everything

- **Hemoglobin:** pocket shaped to bind an oxygen molecule
- Antibodies: Y-shaped to grab specific invaders
- Ion channels: gates that open/close to pass signals through neurons
- Wrong shape → disease (sickle cell, Alzheimer's, prion diseases)
- If you can predict shape from sequence → understand function, design drugs

Hemoglobin, with hemes in red.



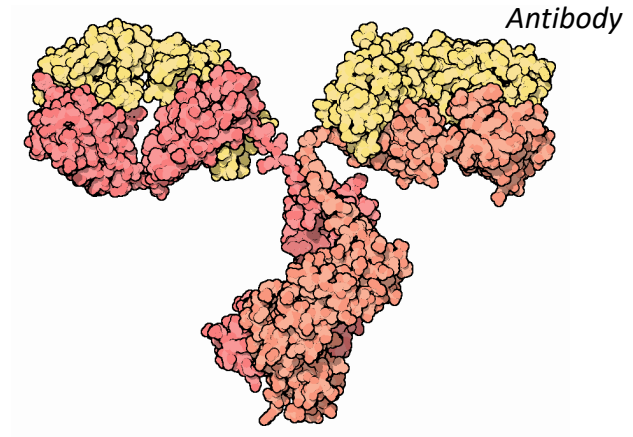
Allosteric motion of hemoglobin, with an oxygen molecule in turquoise.



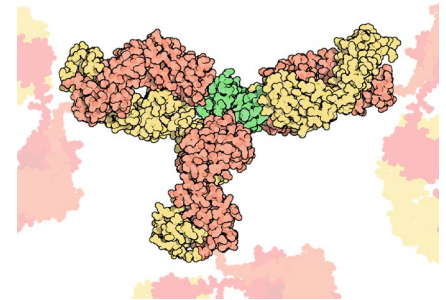
oxy

Structure = Function: Shape Is Everything

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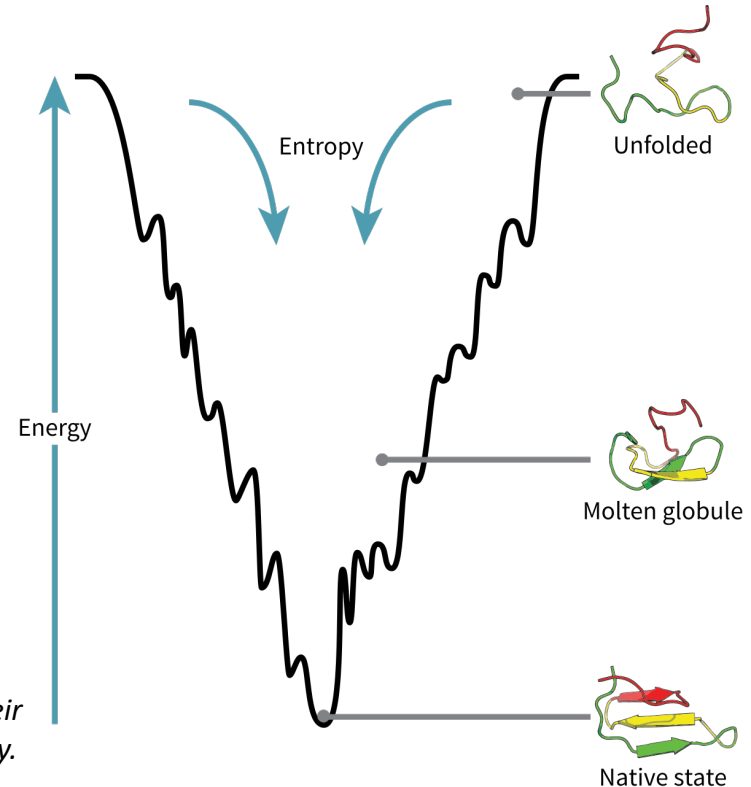


Three antibody Fab fragments that bind to lysozyme (green).



The Protein Folding Problem (Levinthal's Paradox)

- 100-residue protein: $\sim 10^{47}$ possible configurations
- Exhaustive search at nanosecond speed $\rightarrow$ 200 $\times$ the age of the universe
- Yet nature folds proteins in milliseconds to seconds
- There must be a '**folding funnel**' — an energy landscape guiding the chain
- AI question: can we learn that funnel from data, not physics simulation?

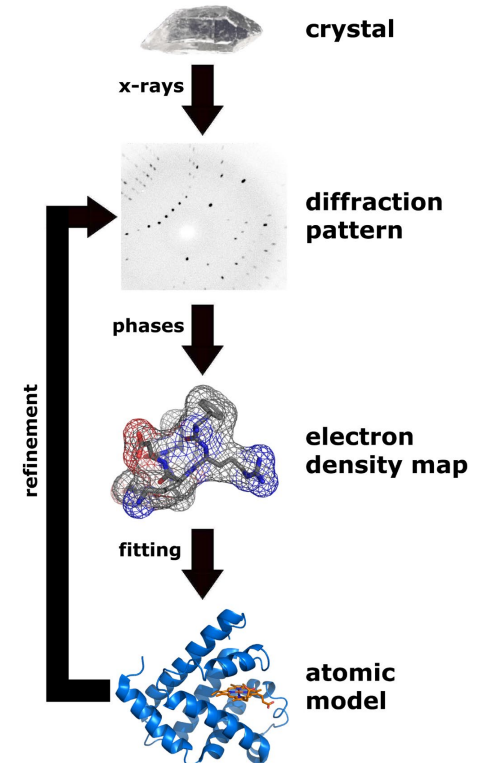


The diagram sketches how proteins fold into their native structures by minimizing their free energy.

Classical Methods: 12 Years and a Chunk of Whale

- Step 1: Crystallize — trick the protein into an ordered crystal (can take years)
- Step 2: Diffract — blast crystal with X-rays → dot pattern
- Step 3: Reverse-engineer — calculate what 3D shape caused that pattern
- John Kendrew: 12 years for myoglobin (1962 Nobel Prize)
- Sourced whale meat from Peru — diving mammals have concentrated myoglobin
- Cost: tens of thousands of dollars per structure
- By 2020: ~150,000 structures total after 60 years of global effort

Structure determination by X-ray crystallography



2024 Nobel Prize in Physics

"for foundational discoveries and inventions that enable machine learning with artificial neural networks"

- **John Hopfield**: energy-based models for information storage (Hopfield networks)
- **Geoffrey Hinton**: backpropagation, Boltzmann machines, deep learning foundations
- Physics concepts → neural network architectures
- These are the ancestors of every architecture we'll discuss today
- Physics gave us the tools; those tools are now solving biology's deepest problems



© Nobel Prize Outreach. Photo:
Nanaka Adachi

John J. Hopfield

Prize share: 1/2



© Nobel Prize Outreach. Photo:
Clément Morin

Geoffrey Hinton

Prize share: 1/2

2024 Nobel Prize in Chemistry

"for computational protein design", the other half jointly to Demis Hassabis and John Jumper "for protein structure prediction"

- **Demis Hassabis & John Jumper** (DeepMind): protein structure prediction (AlphaFold)
- **David Baker** (U. Washington): computational protein design — creating proteins that never existed
- Forward problem: sequence → structure (AlphaFold)
- Inverse problem: desired structure → what sequence? (Rosetta / Baker)
- 2003: Baker designed Top7 — a protein that doesn't exist in nature; it folded as predicted
- Together: the engineering loop is closed



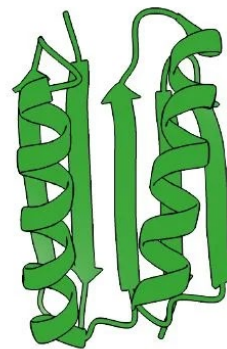
© Nobel Prize Outreach. Photo: Clément Morin
David Baker
Prize share: 1/2



© Nobel Prize Outreach. Photo: Clément Morin
Demis Hassabis
Prize share: 1/4



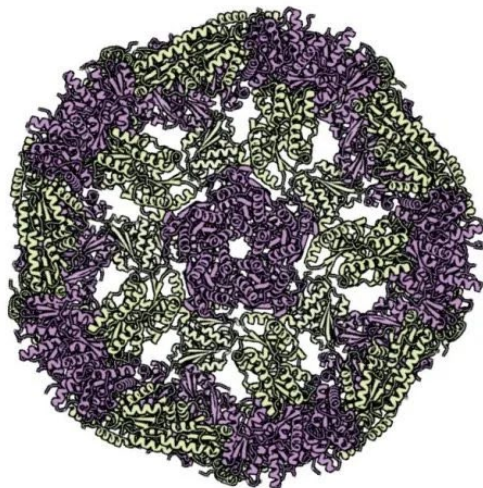
© Nobel Prize Outreach. Photo: Clément Morin
John Jumper
Prize share: 1/4



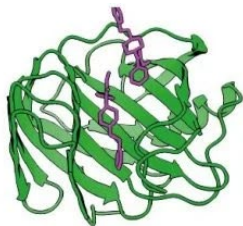
Top7 – the first protein that was entirely different to all known existing proteins. ©Terezia Kovalova/The Royal Swedish Academy of Sciences

2024 Nobel Prize in Chemistry

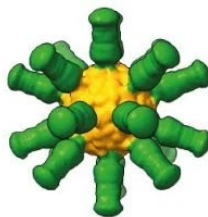
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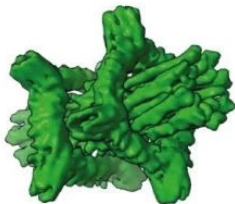
2016: New nanomaterials where up to 120 proteins spontaneously link together.



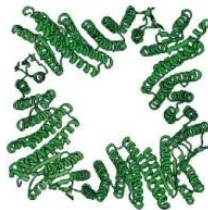
2017: Proteins that bind to an opioid called fentanyl (purple). These could be used to detect fentanyl in the environment.



2021: Nanoparticles (yellow) with proteins imitating influenza virus on the surface (green) that can be used as a vaccine for influenza. Successful in animal models.



2022: Proteins that function as a type of molecular rotor.



2024: Geometrically shaped proteins that can change their shape due to external influences. Could be used for producing tiny sensors.



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David Baker

Prize share: 1/2



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Demis Hassabis

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John Jumper

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Proteins developed using Baker's program *Rosetta*.

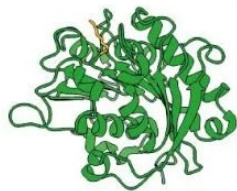
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2024 Nobel Prize in Chemistry

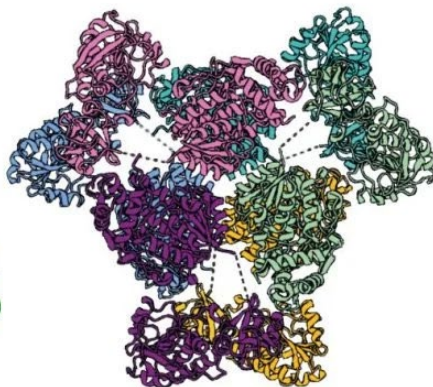
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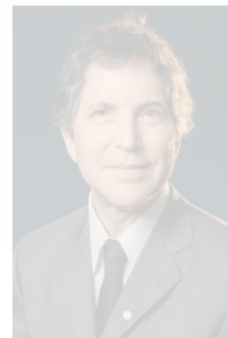
2022: Part of a huge molecular structure in the human body. More than a thousand proteins form a pore through the membrane surrounding the cell nucleus.



2022: Natural enzymes that can decompose plastic. The aim is to design proteins that can be used to recycle plastic.



2023: A bacterial enzyme that causes antibiotic resistance. The structure is important for discovering ways of preventing antibiotic resistance.



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David Baker
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Demis Hassabis
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Protein structures determined using AlphaFold2.

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Part 2: From CNNs to Transformers

The ML evolution:
why CNNs fail for proteins, how attention works, and the bridge to biology

1:30 – 2:00 PM

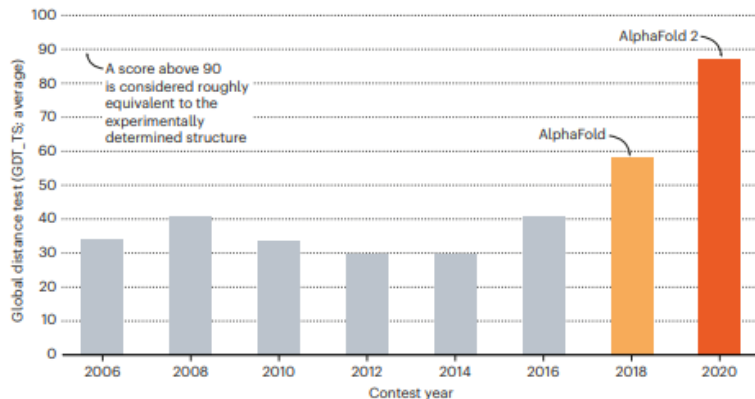
AlphaFold 1: The CNN Approach (2018)

- Used standard CNNs (same as computer vision) to predict a 2D distogram
- **Distogram**: heatmap of predicted distances between all residue pairs
- A separate optimization algorithm then 'folded' the chain to match distances
- Won CASP13: score jumped from ~40 (plateau) to ~70
- But 'solved' threshold = 90 — not there yet
- Key limitation: two-step process (predict distances, then fold separately)

Average GDT tests of the contest winners.

STRUCTURE SOLVER

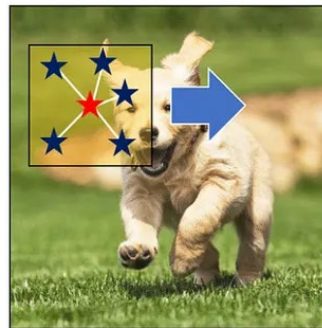
DeepMind's AlphaFold 2 algorithm significantly outperformed other teams at the CASP14 protein-folding contest — and its previous version's performance at the last CASP.



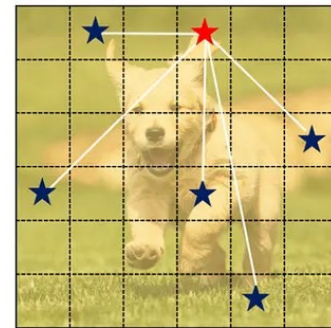
Why CNNs Aren't Enough

- CNNs use local kernels (3×3 , 5×5) — great for nearby patterns
- Receptive field grows slowly (logarithmically) with depth
- Problem: residue 5 and residue 500 can touch in 3D while far apart in 1D
- CNN literally cannot 'see' these long-range contacts efficiently
- We need: every position attending to every other position in one step
- → The Transformer architecture

 Receptive Field



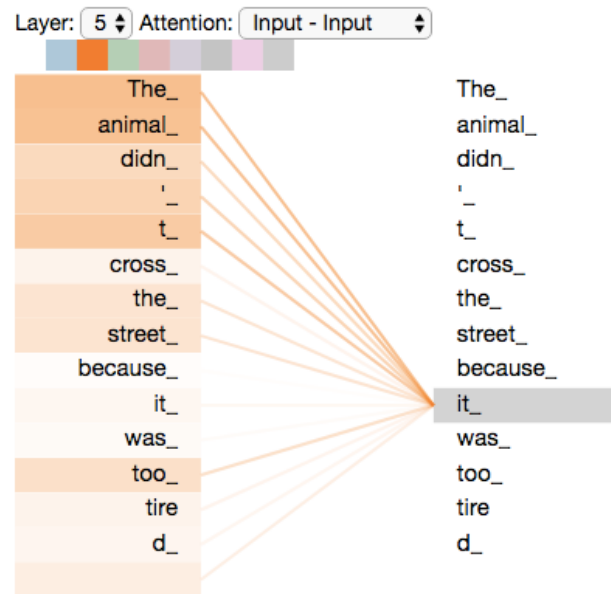
Convolution of CNN



Attention of Vision Transformer

Transformer Intuition: The Sentence Analogy

- "The animal didn't cross the street because it was too tired."
- What does 'it' refer to? The model uses attention to check every other word
- 'tired' → 'it' connects to 'animal'
- "The animal didn't cross the street because it was too wide."
- Change 'tired' to 'wide' → 'it' connects to 'street'
- Key insight: a token's meaning depends on context — not just position
- **Self-attention:** every token looks at every other token and collects context



Self-Attention: Queries, Keys, and Values

- Each token generates 3 vectors:
 - **Query** (what am I looking for?),
 - **Key** (what do I contain?),
 - **Value** (what info do I carry?)
- **Attention score** = dot product of **Query** × **Key** → **relevance**
- Normalize with **softmax** → weights that sum to 1
- **Output** = weighted sum of Values (blend of info from all tokens)
- Formula: **Attention(Q,K,V) = softmax(QK^T / √d) · V**
- All computed in parallel (not sequentially like RNNs)

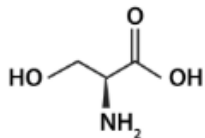
Amino acid 5 Amino acid 480
Q: "hydrophobic?" → K: "yes, hydrophobic" → HIGH score
V: "I'm in the core" → gets mixed in

Amino acid 5 Amino acid 200
Q: "hydrophobic?" → K: "no, I'm charged" → LOW score
V: "I'm on the surface" → mostly ignored

From Words to Amino Acids

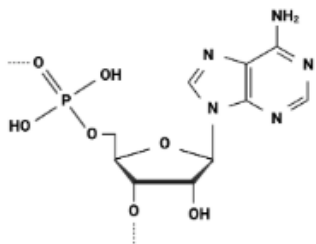
- NLP: word → **token** → embedding vector (semantic meaning)
- Bio-AI: amino acid → **token** → embedding vector (chemical properties)

Standard Amino Acid
(serine)



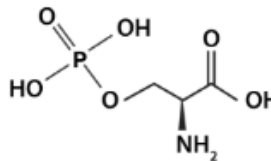
7 atoms*
1 tokens

Standard Nucleotide
(adenosine)



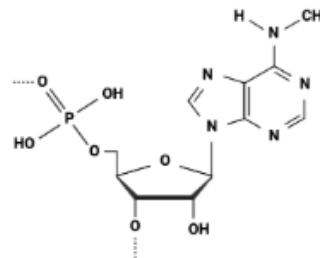
23 atoms*
1 token

Modified Amino Acid
(phosphoserine)



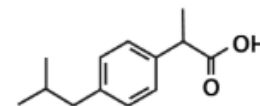
11 atoms*
11 tokens

Modified Nucleotide
(methylenadenosine)



24 atoms*
24 tokens

Ligand
(ibuprofen)

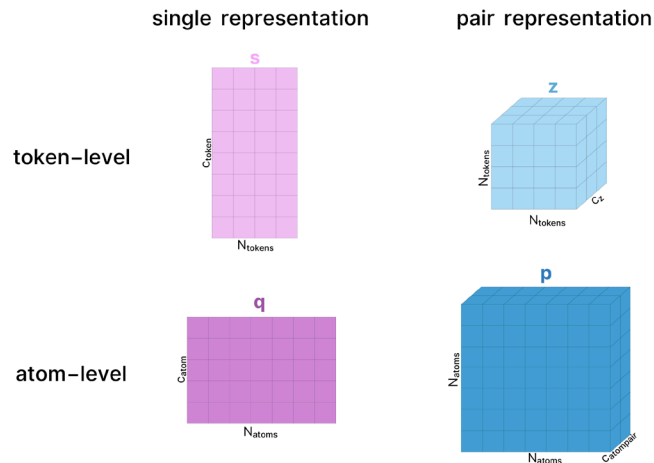


15 atoms*
15 tokens

*atoms=heavy atoms

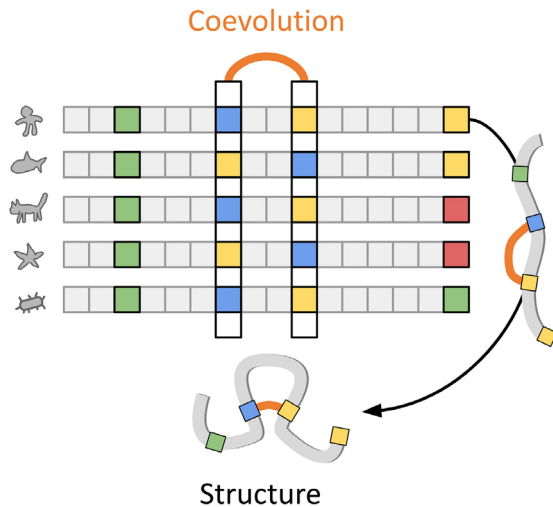
From Words to Amino Acids

- NLP: word → **token** → embedding vector (semantic meaning)
- Bio-AI: amino acid → **token** → embedding vector (chemical properties)
- **Self-attention**: every residue checks relationship to every other residue
- AlphaFold keeps **TWO representations**:
 - Single representation: per residue (what is this residue?)
 - Pair representation: per residue pair (how are i and j related?)
- **The pair representation** = the model's evolving structural hypothesis



Evolution as a Teacher: Multiple Sequence Alignments

- Compare the same protein across hundreds of species
- If residue 23 mutates, and residue 187 always co-mutates → touching in 3D
- This is co-evolution: nature's built-in signal for 3D contacts
- MSAs are the primary input to AlphaFold (alongside templates)
- Richer MSA → better prediction; <30 sequences → accuracy drops
- AlphaFold searches databases of billions of sequences



Multi-Head Attention: Multiple Perspectives

- Single attention head: one 'question' per position
- Multi-head: run 8–32 heads in parallel, each with different Q/K/V projections
- Each head specializes: local contacts, long-range, chemistry, co-evolution...
- Outputs concatenated and linearly projected
- AlphaFold 2's **Evoformer** uses multi-head attention extensively

Positional Encoding & SE(3) Symmetry

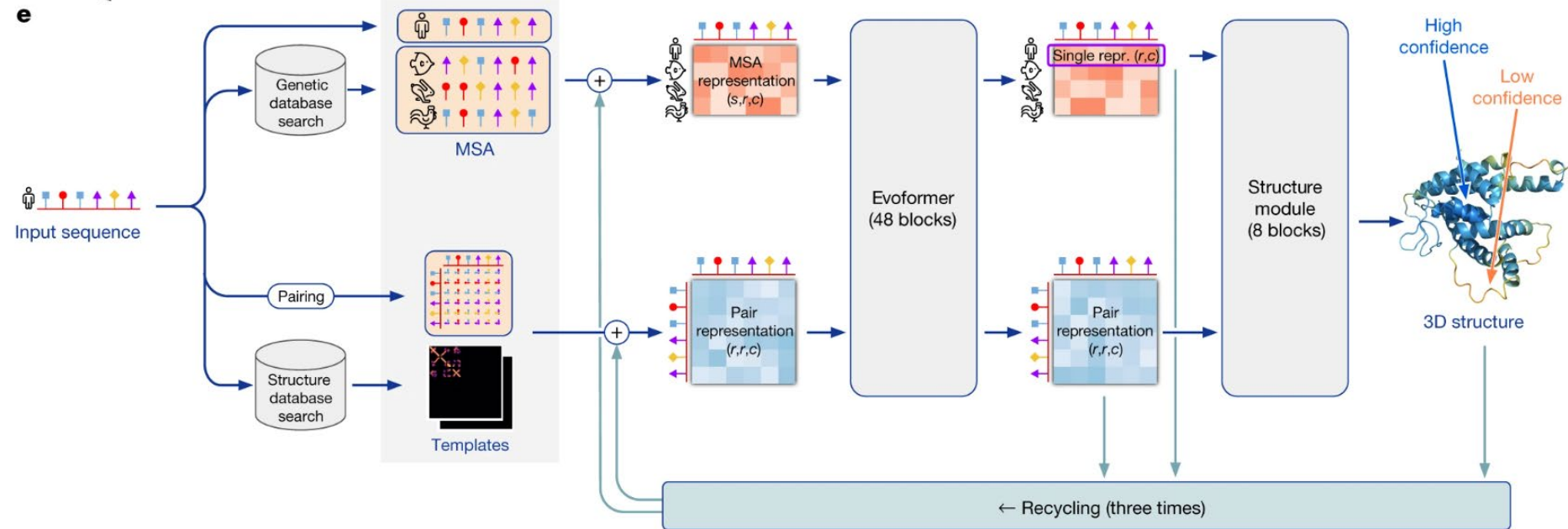
- Attention is permutation-invariant — doesn't know token order by default
- Solution: add position-dependent signal to each embedding
- NLP: sinusoidal functions (Vaswani et al. 2017)
- AlphaFold: relative **positional encodings** in the pair representation
- Deeper point: protein structures must be rotation/translation invariant
- AF2's structure module uses **SE(3) equivariance** via **Invariant Point Attention**
 - Physics built into the architecture

Part 3: AlphaFold Architecture Dissected

AF2's Evoformer, Structure Module, Recycling — then the AF3 paradigm shift

2:00 – 2:30 PM

AlphaFold 2: The Big Picture



- Inputs: MSA + structural templates + raw sequence
- Trunk: The Evoformer — 48 identical blocks processing two streams:
 - MSA representation ($N_{\text{seq}} \times N_{\text{res}}$)
 - Pair representation ($N_{\text{res}} \times N_{\text{res}}$)
- Structure Module: converts representations $\rightarrow$ 3D atom coordinates
- Recycling: feed output back into trunk 3 times (4 total passes)
- Trained on $\sim 170,000$ experimental structures from the PDB

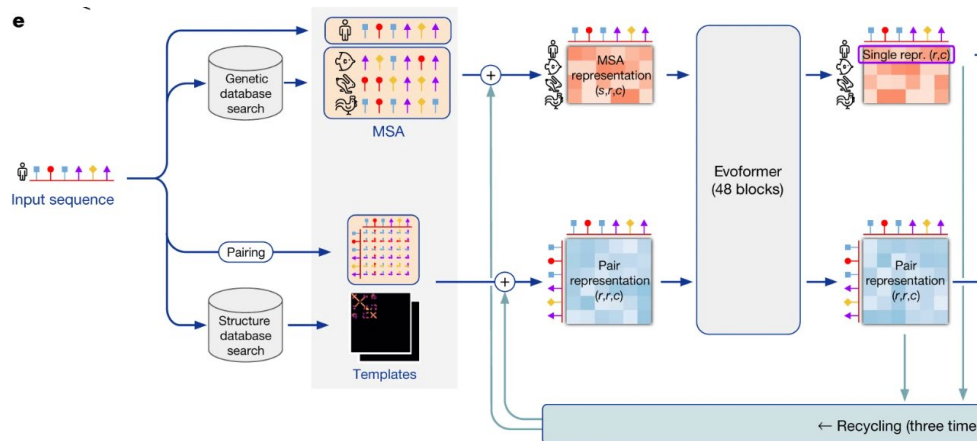
Slide 19 — The Evoformer: Dual Towers

MSA Tower (Biology):

- Row-wise attention: each position attends to all other positions $\rightarrow$ per-residue conservation
- Column-wise attention: each position across all species $\rightarrow$ co-evolution signal

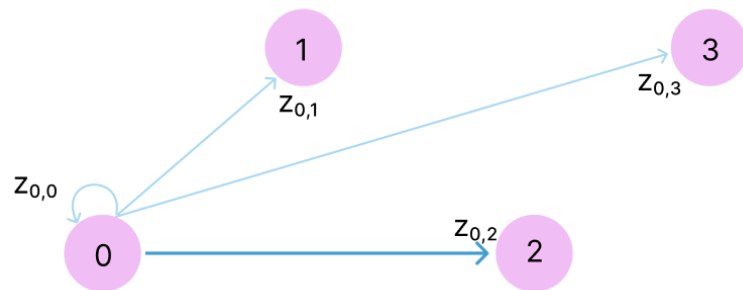
Pair Tower (Geometry):

- $N_{\text{res}} \times N_{\text{res}}$ matrix of pairwise relationships
- Updated by: outer product of MSA + triangular updates
- Geometry biases MSA attention back $\rightarrow$ cross-talk $\times 48$



Triangular Attention & Updates

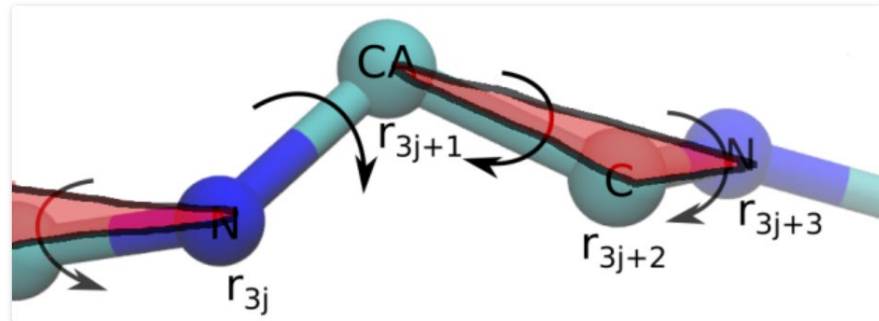
- Pair representation must be geometrically consistent
- Triangle inequality: $d(A,B) \leq d(A,C) + d(C,B)$
- Triangular multiplicative update: for every pair (i,j) , check all third residues k
 - 'Starting node': uses $z_{\{i,k\}}$ and $z_{\{j,k\}} \rightarrow$ update $z_{\{i,j\}}$
 - 'Ending node': uses $z_{\{k,i\}}$ and $z_{\{k,j\}} \rightarrow$ update $z_{\{i,j\}}$
- Triangular self-attention: attention biased by the third edge
- Not a hard constraint — strongly encourages consistency through architecture



Structure Module: Residue Gas $\rightarrow$ 3D Coordinates

- Initialize each residue as a free-floating rigid body ('frame') at the origin
- Each frame: position (x,y,z) + rotation (3x3 matrix) = local coordinate system
- **Invariant Point Attention (IPA)** moves these frames iteratively
- IPA is SE(3) equivariant: invariant to global rotation and translation
- After 8 layers: frames settle into physically valid backbone positions
- Side-chain torsion angles predicted separately

The structure module considers the protein as a “residue gas”. Every amino acid is modelled as a triangle, representing the three atoms of the backbone. These triangles float around in space, and are moved by the network to form the structure.



The “residue gas” approach. Image taken from the [OpenFold 2 webpage](https://www.blogig.com/blog/2021/07/alphafold-2-is-here-whats-behind-the-structure-prediction-miracle/), by Georgy Derevyanko.

Structure Module: Residue Gas → 3D Coordinates

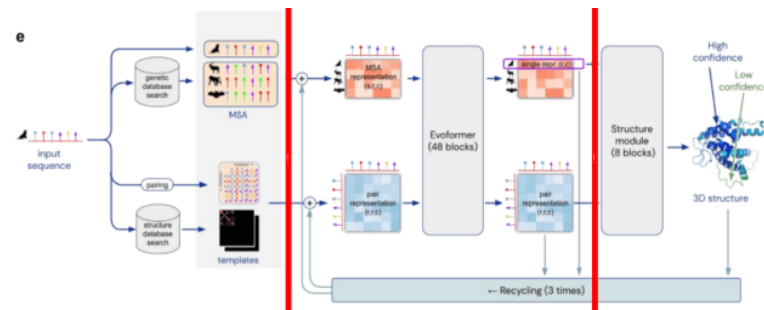


Recycling iteration 0, block 01
Secondary structure assigned from the final prediction

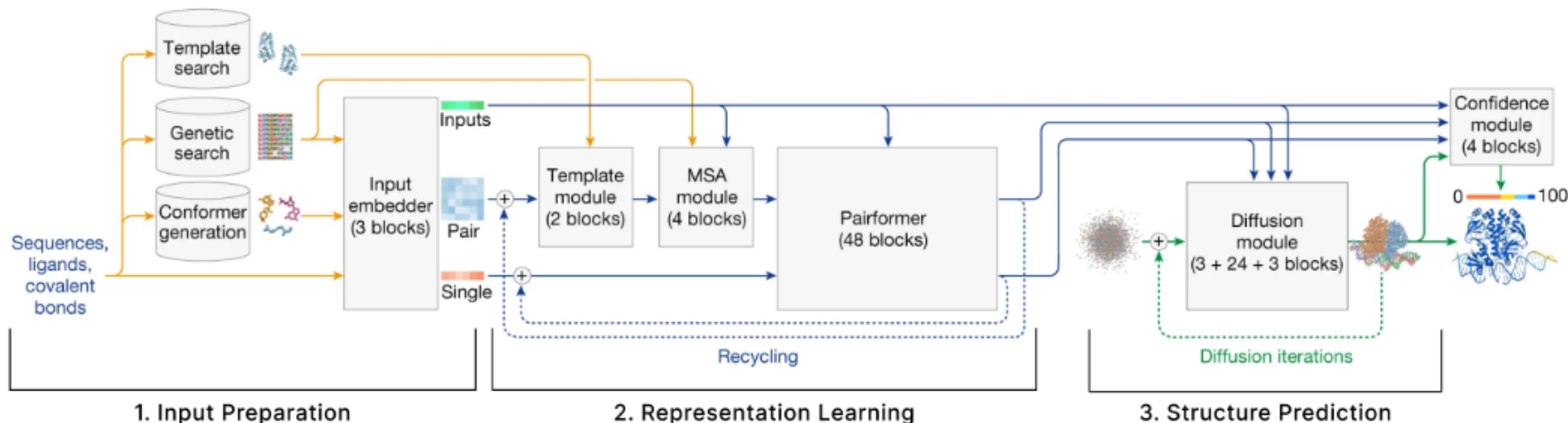
Recycling: Iterative Refinement

- Output 3D structure fed back into the Evoformer as additional pair information
- Happens 3 times (4 total passes through the network)
- Each pass refines the structure — like successive drafts of a paper
- Some errors only become apparent once you have a 3D hypothesis
- CASP14 result: median GDT-TS ~92

→ The 'protein folding problem' declared effectively solved for single chains



AlphaFold 3: What Changed (2024)



- Evoformer → Pairformer: MSA simplified, info bottlenecks into Pair representation
- **Structure Module** → Diffusion Module: no rigid frames; random atom cloud → denoise
- **Why diffusion?** Ligands and ions don't have 'frames' or torsion angles
- AF2 = physicist's approach (rigid bodies); AF3 = sculptor's approach (chisel from marble)
- 50% improvement in protein-ligand accuracy vs. traditional docking
- Note: diffusion theory covered in depth in Week 11

AlphaFold 3: What Changed (2024)



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The Illustrated AlphaFold

A visual walkthrough of the AlphaFold3 architecture, with more details and diagrams than you were probably looking for.

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PUBLISHED

July 10, 2024

AlphaFold 3: What Changed (2024)



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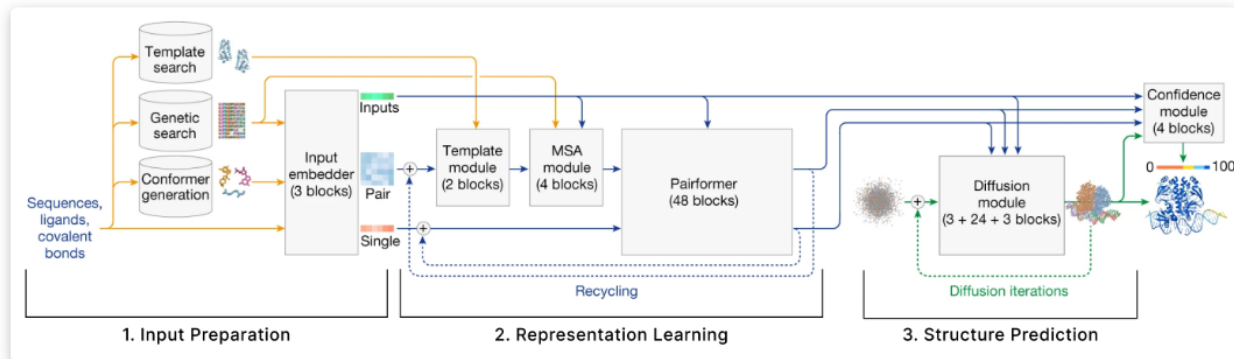
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A visual v
than you

AUTHORS
Elana Simon
Jake Silberg

Tokenization is described in its own section, but for now just know that when we say “token” it either represents a single amino acid (for proteins), nucleotide (for DNA/RNA), or an individual atom if that atom is not part of a standard amino acid/nucleotide.

Interactive Table of Contents



Throughout the post, we highlight where you are in this diagram so you don't get lost!

Full architecture. If you click on any part of the architecture, it will take you to that section of the post. If you resize the page, you might need to refresh to keep the interactive part working. (Diagram modified from AF3 paper)

AF2 vs. AF3: Side-by-Side

Feature	AlphaFold 2 (2020)	AlphaFold 3 (2024)
Input molecules	Proteins only	Proteins, DNA, RNA, ligands, ions
Trunk	Evoformer (48 blocks)	Pairformer (48 blocks)
MSA usage	Heavy (row + column attn)	Simplified, de-emphasized
Structure module	Rigid frames + IPA + torsion	Diffusion (atom-level denoising)
Symmetry	SE(3) via IPA	SE(3) via diffusion conditioning
Key metric	GDT-TS ~92 (CASP14)	+50% on protein-ligand docking

The Model Landscape: Beyond AlphaFold

Model	Architecture	Molecules	License	Key Feature
AlphaFold 3	Pairformer + Diffusion	All	Academic	Gold standard accuracy
Boltz-2	AF3-like	All + affinity	MIT (open)	Binding affinity, 1000× faster than FEP
Chai-1	AF3-like	All	Acad/Comm	ESM embedding mode (no MSA)
ESMFold	pLM (no MSA)	Proteins	Open	Single-sequence, very fast
RoseTTAFold AA	3-track	All	Academic	Baker Lab alternative



Break

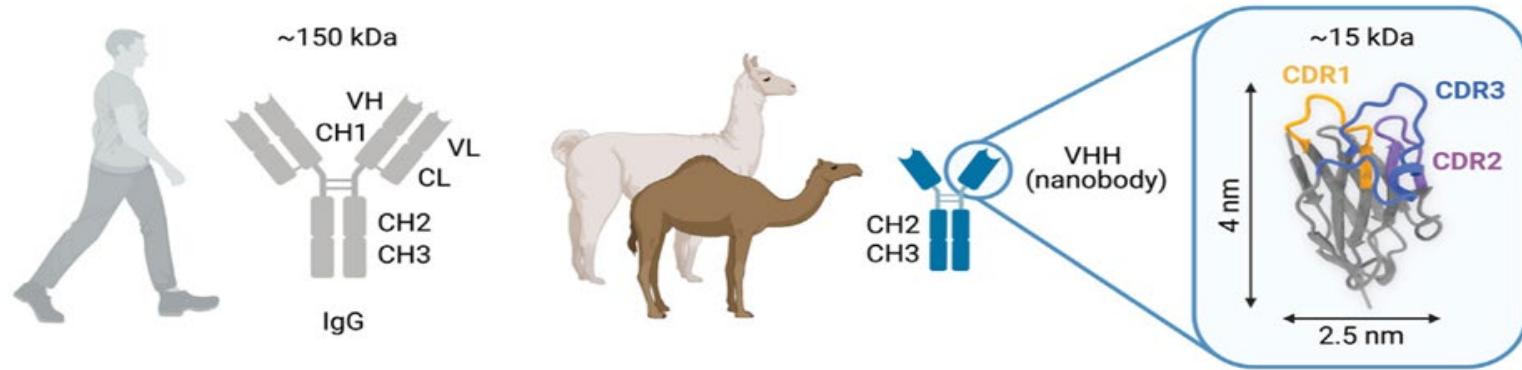
2:30 – 2:50 PM

Part 4: Benchmarking Case Study

Can AI distinguish real binders from fakes? Results from nanobody-antigen benchmarking

2:50 – 3:15 PM

The Background – What are Nanobodies

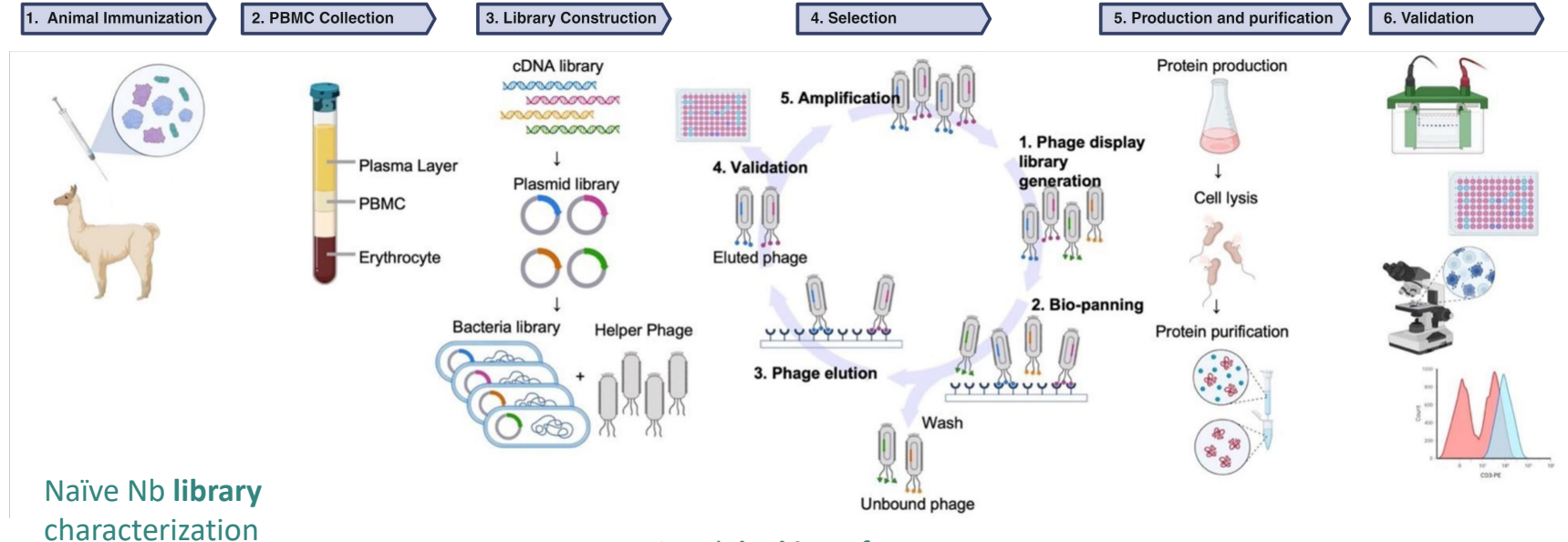


Key advantages:

- **Small** → reach cryptic epitopes, better tissue penetration
- **Stable** → harsh conditions, long shelf life
- **Cost-effective** → microbial production (E. coli)
- **Versatile** → engineering, multimerization, intracellular delivery
- **Clinically validated** → Caplacizumab approved for thrombosis

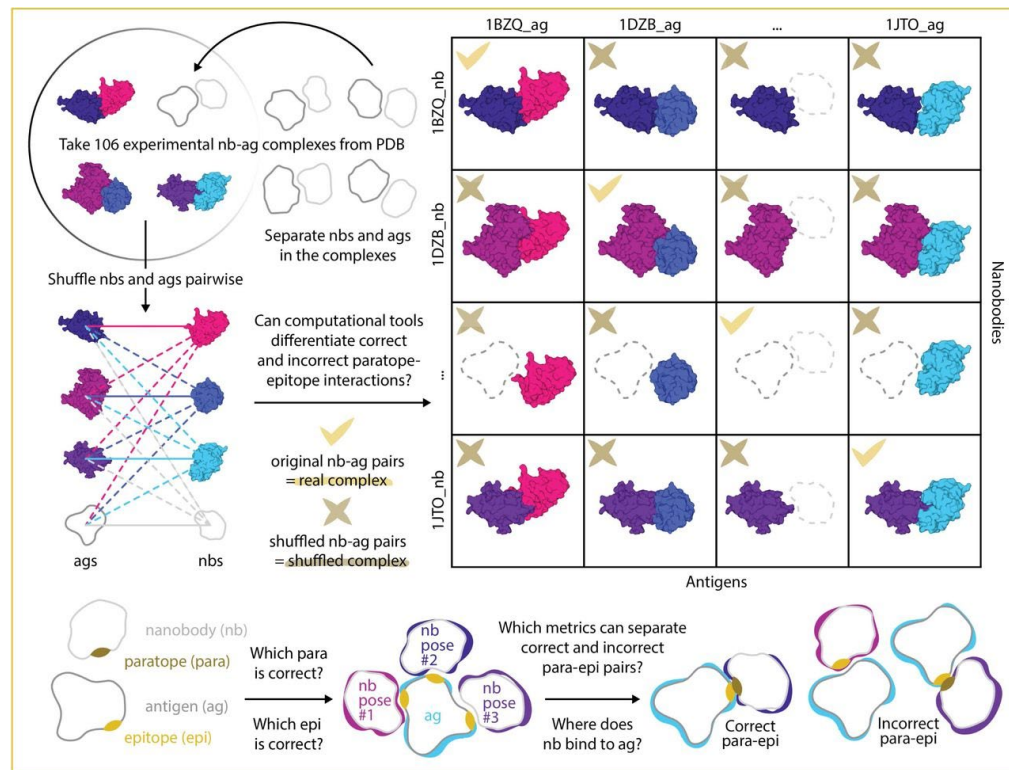
The Background – Nanobodies Discovery Pipeline

Can *in silico* selection replace parts of traditional *in vitro* screening?



The Benchmarking Challenge

- System: Nanobody-Antigen (VHH-Ag) interactions — key for drug discovery
- Ground truth: 106 experimentally validated complexes from the PDB
- Negative controls: 11,342 'shuffled' pairs — randomly mismatched (known non-binders)
- Tools tested: AlphaFold 3, Boltz-2, Chai-1
- The question: Do models assign higher confidence to real binders than decoys?



The Benchmarking Challenge

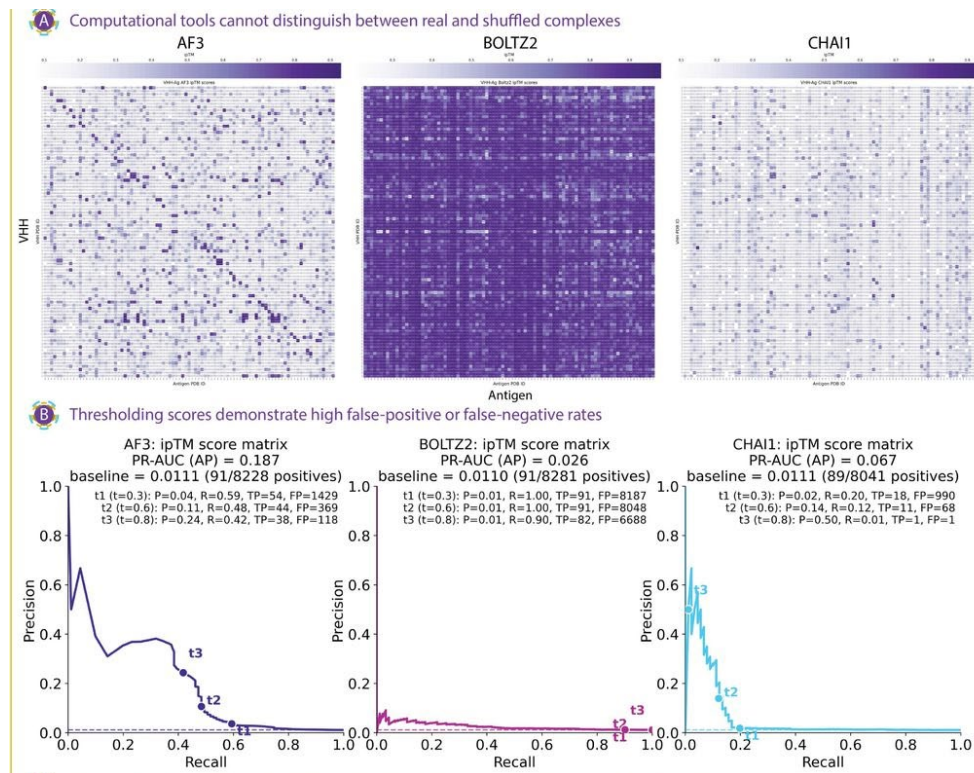
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***8E0E**; inhibition of Ca V beta 1 -associated voltage-dependent calcium channels with a functionalized nanobody*

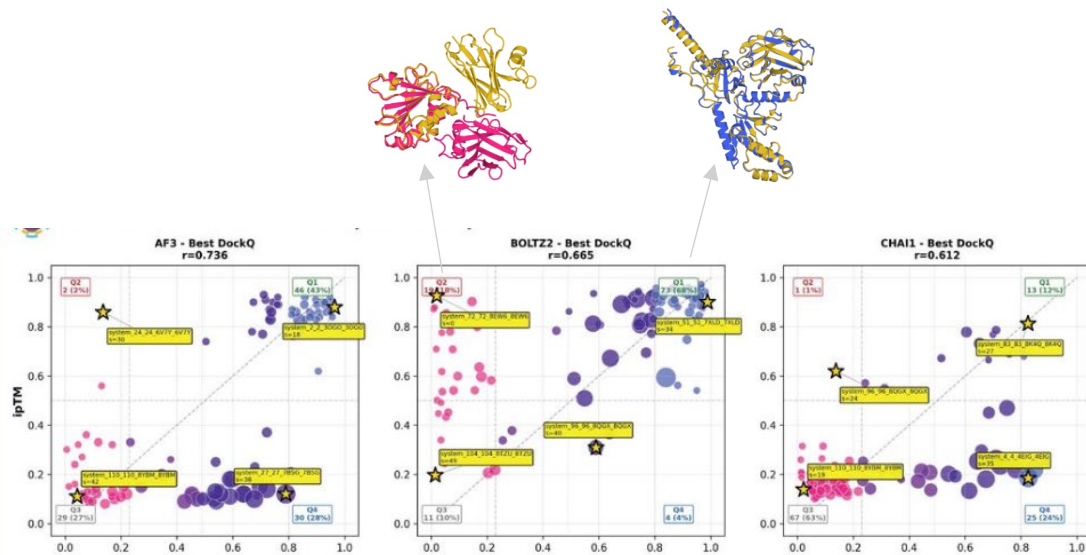
Finding 1: The Specificity Blind Spot

- Precision-Recall curves: all tools track near the random baseline
- Models generate 'geometrically polished' interfaces for non-binding pairs
- No steric clashes, plausible hydrogen bonds — but biologically meaningless
- AI learned what interfaces 'look like' — not what makes a specific pair bind
- Critical warning for drug discovery: *enormous false positive risk*



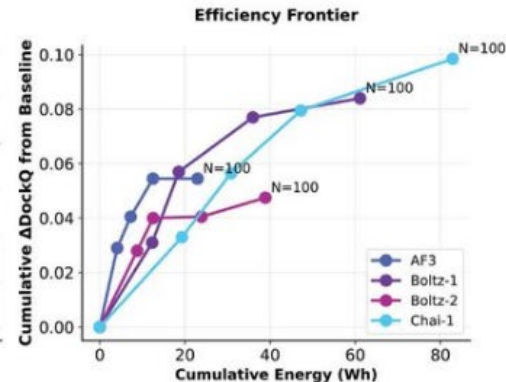
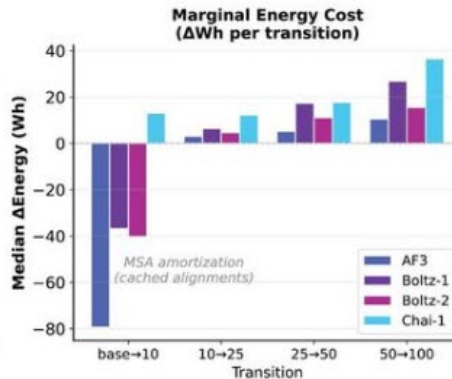
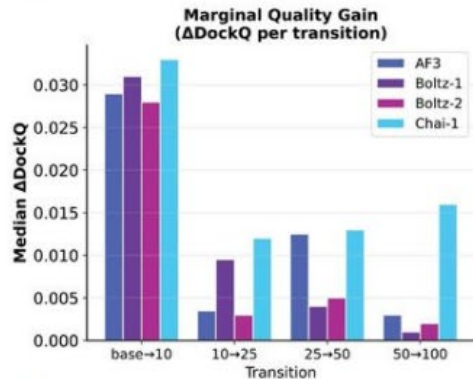
Finding 2: The Confidence Disconnect

- ipTM (model's confidence) does NOT correlate with DockQ (actual quality)
- Boltz-2: systematically overconfident (high ipTM even for failures)
- Chai-1: systematically underconfident (low ipTM even for successes)
- Scores are 'locked-in' to trunk's first hypothesis — don't update with sampling
- ipTM itself is not reliable to score probability of binding



Finding 3: Sampling Efficiency

- More samples help, but with sharply diminishing returns
- Most quality gains captured in N=10–25 samples
- At N=100: Chai-1 uses 82.9 Wh; AF3 uses 23.0 Wh per complex
- AF3: high fixed MSA cost, low marginal cost per sample
- Chai-1 (embedding): low startup, but steep linear scaling
- Recommendation: staged workflow — N=10 first, deep sampling for high-value targets only



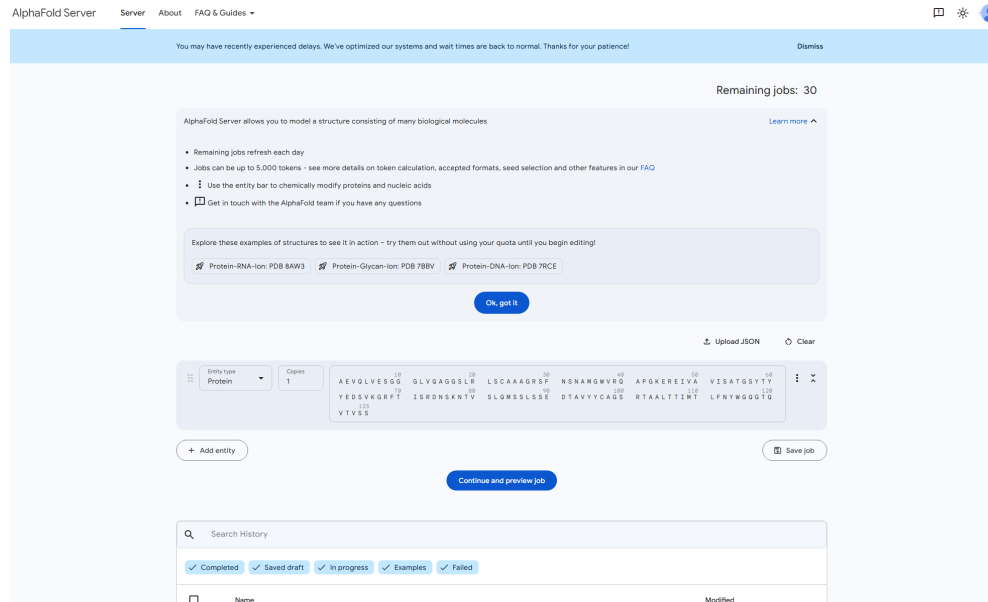
Part 5: Practical Guide

How to use these tools, read the diagnostics, and spot failures

3:15 – 4:00 PM

Accessing the Tools

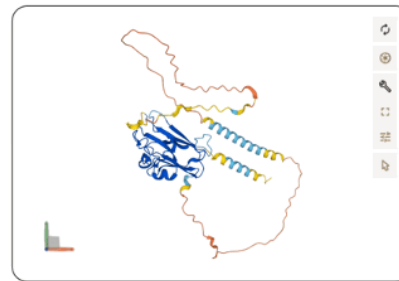
- AlphaFold Server: free web interface, 30 jobs/day → alphafoldserver.com
- Boltz-2: fully open-source (MIT), Colab or HPC → github.com/jwohlwend/boltz
- Chai-1: open-source, embedding mode → github.com/chaidiscovery/chai-lab
- ColabFold: AF2 on Google Colab, fast MSA → github.com/sokrypton/ColabFold
- For HPC/SLURM: Boltz-2 and AF3 support GPU-accelerated batch submission



The Confidence Scores: pLDDT, ipTM, PAE

- **pLDDT (predicted Local Distance Difference Test; 0-100):** per-residue local confidence
 - >90 = high accuracy (dark blue) | <50 = disordered/hallucinated (orange)
- **ipTM (Predicted Aligned Error; 0-1):** interface/complex confidence
 - >0.8 = high quality | <0.6 = extreme caution | Not well-calibrated
- **PAE (Predicted Aligned Error):** 2D heatmap (N_res × N_res)
 - Dark sharp blocks = confident positioning | Fuzzy = uncertain
 - Cross-chain blocks show interface quality

Ephrin-B2



Model Confidence ⓘ

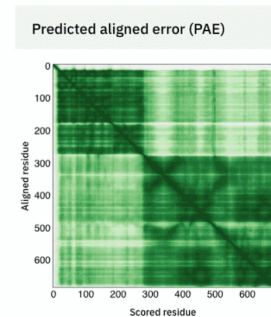
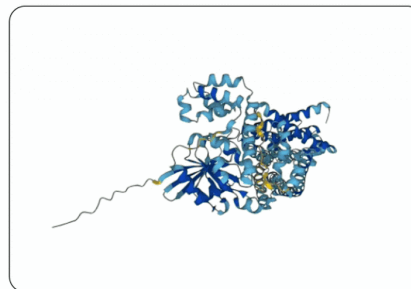
Very high (pLDDT > 90)	Low (70 > pLDDT > 50)
High (90 > pLDDT > 70)	Very low (pLDDT < 50)

Sample **pLDDT** for a predicted protein structure. Different colours indicate AlphaFold2's level of confidence in its prediction (see key).

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Ferrous iron transport protein B



Model Confidence

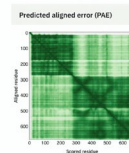
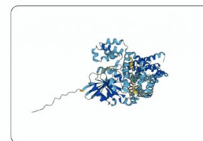
- | | |
|--------------------------|-------------------------|
| ■ Very high (pLDDT > 90) | ■ Low (70 > pLDDT > 50) |
| ■ High (90 > pLDDT > 70) | ■ Very low (pLDDT < 50) |

A sample PAE plot, showing regions of high confidence (dark green) and low confidence (pale green) for the predicted structure of ferrous iron transport protein B (AlphaFold ID: [AF-Q12XU1-F1](#)).

The Confidence Scores: pLDDT, ipTM, PAE

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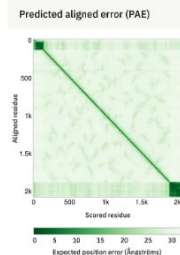
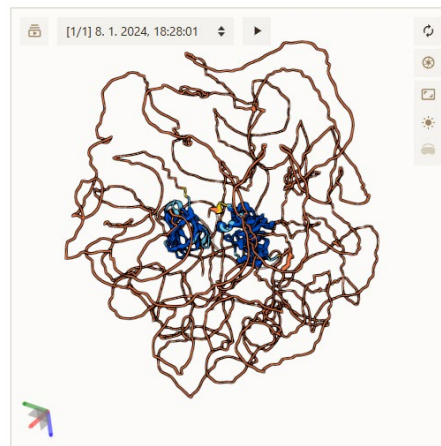
Ferrous iron transport protein B



Model Confidence

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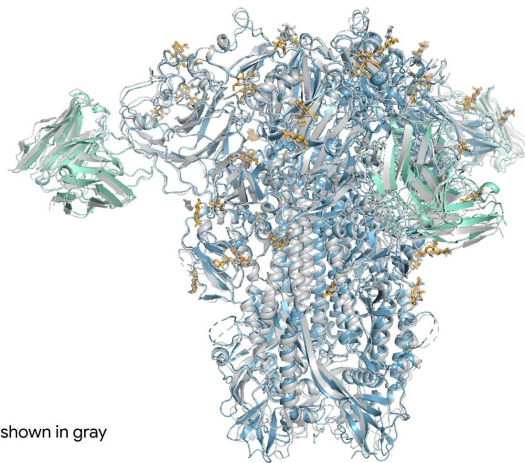


Mediator of DNA damage checkpoint protein 1

Case 1: The Structural Success (PDB 7PNM)

- System: OC43 coronavirus spike + antibodies + glycans (4,665 residues)
- pLDDT: >90 throughout (dark blue)
- ipTM: high
- PAE: sharp blocks at every interface
- RMSD to experiment: ~ 0.8 Å, a near-experimental accuracy
- VERDICT: TRUST. When all three metrics agree, the prediction is reliable.

7PNM



Ground truth shown in gray

Spike protein of a common cold virus (Coronavirus OC43)

Case 2: The Confident Decoy

- System: shuffled pair — nanobody + mismatched antigen (never binds)
- Boltz-2 assigns ipTM = 0.85, generates clash-free interface
- Geometry looks perfect — but biologically impossible
- Model predicts 'what interfaces look like' — not whether THIS one exists
- VERDICT: DECOY.
- Lesson: always include negative controls (shuffled pairs) alongside real predictions



6k69 pdb vs 8ukv pdb

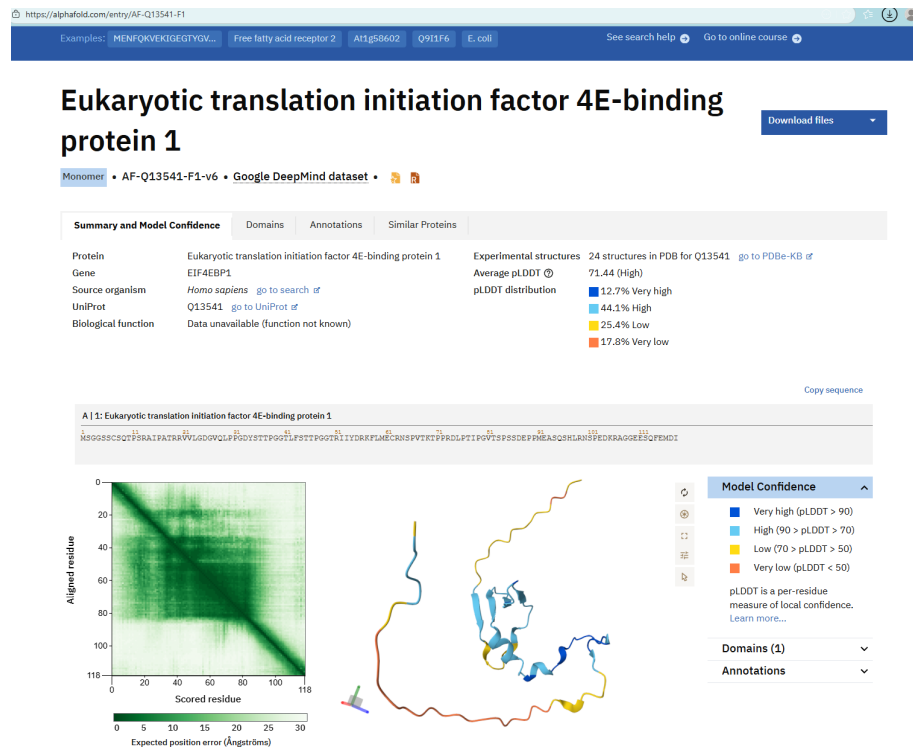
Case 3: Hallucinating Order

- System: protein with long **intrinsically disordered regions** (IDRs, database: <https://disprot.org/>)
- AF3 'hallucinates' beautiful alpha-helices in floppy regions
- 3D rendering looks convincing — but pLDDT < 50 (orange/yellow)
- The model is telling you it's guessing — trust the scores, not the picture
- VERDICT: HALLUCINATION.
- Any region with pLDDT < 50 should be treated as unreliable

The screenshot shows the DisProt website homepage. At the top is a navigation bar with links: Browse, Training, Ontology, Release notes, Download, Deposition, About, and Help. Below the navigation bar is a welcome message: "Welcome to DisProt, the database of intrinsically disordered proteins". A search bar is present with the text "Search in DisProt" and a "Search" button. Below the search bar are examples of search terms: VHL, HMGA1, and Nucleoprotein. The main content area is divided into two columns: "Organisms" and "Datasets". The "Organisms" column lists various species with their corresponding protein counts: H. sapiens (1,301), M. musculus (201), R. norvegicus (89), S. cerevisiae (214), E. coli (8), A. thaliana (113), D. melanogaster (64), and C. elegans (56). The "Datasets" column lists various protein categories with their corresponding protein counts: Bacterial virulence-related proteins (76), Age-related disorders proteins (42), RNA-binding proteins (212), Condensates-related proteins (181), NDDs-related proteins (324), Neglected tropical diseases proteins (105), Autophagy-related proteins (109), Cancer-related proteins (181), Viral proteins (230), Extracellular matrix proteins (100), and Unicellular toxins and antitoxins (51). Below the main content area are three sections: "DisTriage link" (a literature triage service), "Latest News" (recent updates and news), and "How to cite" (information on how to cite the database). At the bottom of the page are logos for UniProt, InterPro, and EGO.

Case 3: Hallucinating Order

- System: protein with long **intrinsically disordered regions** (IDRs, database: <https://disprot.org/>)
- AF3 'hallucinates' beautiful parts of the structure in floppy regions
- 3D rendering looks convincing — but pLDDT < 50 (orange/yellow)
- The model is telling you it's guessing — trust the scores, not the picture
- Any region with pLDDT < 50 should be treated as unreliable



Takeaway: Staged & Multi-Tool Workflow

1. SCREEN: Run N=10 samples across multiple tools (AF3 + Boltz-2)
2. FILTER: Check for cross-tool agreement on binding mode
3. VALIDATE: Always include shuffled/negative controls
4. REFINE: Deep sampling (N=50–100) only for high-value hits
5. VERIFY: Experimental validation remains the ground truth

AI generates hypotheses, not proofs. Cross-tool agreement > any single score.

Resources & Learning Path

Tools

AlphaFold Server: alphafoldserver.com

Boltz-2: github.com/jwohlwend/boltz (MIT license)

Chai-1: github.com/chaidiscovery/chai-lab

ColabFold: github.com/sokrypton/ColabFold

Learning

EMBL-EBI AlphaFold Training: ebi.ac.uk/training/online/courses/alphafold/

The Illustrated AlphaFold (Simon & Silberg, ICLR 2025): elanapearl.github.io/blog/2024/the-illustrated-alphafold/

The Illustrated Transformer (Alammar): jalamar.github.io/illustrated-transformer/

BLOPIG Oxford Blog (AF2 + AF3): blopig.com/blog/tag/alphafold/

Key Papers

Jumper et al. (2021) Nature — AlphaFold 2: doi.org/10.1038/s41586-021-03819-2

Abramson et al. (2024) Nature — AlphaFold 3: doi.org/10.1038/s41586-024-07487-w

Vaswani et al. (2017) — Attention Is All You Need: arxiv.org/abs/1706.03762

Smorodina, Ali, Kropivšek et al. (2025) — VHH-Ag Benchmarking

Appendix: Q&A Preparation

Q&A — Architecture & Theory

Q: Why 48 Evoformer blocks?

A: Empirically determined. More blocks help but with diminishing returns on accuracy vs. compute.

Q: What's the loss function?

A: Multi-task: FAPE on backbone atoms + auxiliary losses (distogram, masked MSA, pLDDT head).

Q: How does AF2 attention differ from standard?

A: Biaxial (row+column on MSA), triangular (constrained by pairs), IPA (SE(3)-equivariant). Standard uses vanilla self-attention.

Q: How does AF3 diffusion work?

A: Denoising diffusion: add noise to true coords at training, start from noise at inference, denoise with Pairformer conditioning. Week 11 covers theory.

Q: Can AlphaFold predict dynamics?

A: Not really. Predicts static ground state. Deep sampling captures some diversity. MD simulations still needed for dynamics.

Q: How does ESMFold work without MSA?

A: ESM is a protein language model (like BERT). Learns implicit evolutionary constraints from millions of sequences. Faster but generally less accurate.

Q&A — Practical & Benchmarking

Q: Why do models fail at binding specificity?

A: Trained to minimize FAPE against PDB structures, not to discriminate binders. No negative examples in training.

Q: How much GPU do I need?

A: ColabFold: free (T4). Boltz-2: 16–40GB VRAM (A100/V100). AF3 full: 40–80GB + CPU for MSA.

Q: Can these predict mutation effects?

A: Poorly. Too subtle for current models. Use EVE or ESM-1v for mutation-effect prediction instead.

Q: Can I use AF3 for protein design?

A: AF3 is forward (seq→struct). For design use RFdiffusion or ProteinMPNN. Use AF3 to validate designs.

Q: How do I know if MSA is deep enough?

A: Check Neff (effective sequences). Below ~30, accuracy drops. ESMFold (no MSA) may be better for orphan proteins.

Q: Energy cost comparison?

A: AF3 MSA: ~82 Wh fixed. Per-sample: ~0.2 Wh. Chai-1 embed: ~6.6 Wh base, ~0.8 Wh/sample. At N=10 all are comparable.